Laboratory Work 3 - Merged Data, DAX, and DAX Function Insights

Data Modeling Fundamentals Using Merged Data

Part 1: Review the Merged Dataset

	OrderID	CustomerID	OrderDate	SalesAmount	CustomerName	City
1	O001	C009	1/4/2025	3000	Customer 9	Cagayan de Oro
2	O109	C009	3/7/2025	2200	Customer 9	Cagayan de Oro
3	O112	C009	1/20/2025	3000	Customer 9	Cagayan de Oro
4	O188	C009	3/18/2025	7500	Customer 9	Cagayan de Oro
5	O173	C001	2/6/2025	3000	Customer 1	Legazpi
6	O002	C156	2/12/2025	7500	Customer 156	Dumaguete
7	O020	C156	1/28/2025	800	Customer 156	Dumaguete
8	O130	C156	3/9/2025	45000	Customer 156	Dumaguete
9	O080	C002	2/2/2025	800	Customer 2	Manila
10	O167	C002	1/4/2025	2200	Customer 2	Manila
11	O003	C137	3/11/2025	800	Customer 137	Davao
12	O032	C137	3/21/2025	1200	Customer 137	Davao
13	O148	C003	1/17/2025	1200	Customer 3	Butuan
14	O004	C127	1/1/2025	45000	Customer 127	Tacloban
15	O028	C127	1/10/2025	800	Customer 127	Tacloban
16	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Guide Questions

1. Is the dataset in flat-table format?

   - Yes, it's a flat table because all columns (OrderID, CustomerID, OrderDate, SalesAmount, CustomerName, City, Region) are in a single table.

2. Which columns describe customers?

   - CustomerID, CustomerName, City,Region

3. Which columns represent transactions?

   - OrderID, OrderDate, SalesAmount

Part 2: Identify Fact and Dimension Tables

Dim Customer

	CustomerID	CustomerName	City	Region
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C001	Customer 1	Legazpi	Luzon
3	C156	Customer 156	Dumaguete	Visayas
4	C002	Customer 2	Manila	Luzon
5	C137	Customer 137	Davao	Mindanao
6	C003	Customer 3	Butuan	Mindanao
7	C127	Customer 127	Tacloban	Visayas
8	C004	Customer 4	Dumaguete	Visayas
9	C105	Customer 105	Butuan	Mindanao
10	C054	Customer 54	Cagayan de Oro	Mindanao
11	C091	Customer 91	Manila	Luzon
12	C007	Customer 7	Zamboanga	Mindanao
13	C016	Customer 16	Butuan	Mindanao
14	C027	Customer 27	Baguio	Luzon
15	C094	Customer 94	Dumaguete	Visayas
16	C010	Customer 10	Cagayan de Oro	Mindanao
17	C146	Customer 146	Tacloban	Visayas
18	C012	Customer 12	Cagayan de Oro	Mindanao
19	C131	Customer 131	Cebu	Visayas
20	C013	Customer 13	Cagayan de Oro	Mindanao
21	C170	Customer 170	Dumaguete	Visayas
22	C014	Customer 14	Baguio	Luzon

Part 2: Identify Fact and Dimension Tables

Fact Sales

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	3000
2	O109	C009	3/7/2025	2200
3	O112	C009	1/20/2025	3000
4	O188	C009	3/18/2025	7500
5	O173	C001	2/6/2025	3000
6	O002	C156	2/12/2025	7500
7	O020	C156	1/28/2025	800
8	O130	C156	3/9/2025	45000
9	O080	C002	2/2/2025	800
10	O167	C002	1/4/2025	2200
11	O003	C137	3/11/2025	800
12	O032	C137	3/21/2025	1200
13	O148	C003	1/17/2025	1200
14	O004	C127	1/1/2025	45000
15	O028	C127	1/10/2025	800
16	O179	C004	1/8/2025	1200
17	O005	C105	3/3/2025	1200
18	O006	C054	3/29/2025	2200
19	O045	C054	3/30/2025	1200
20	O007	C091	2/7/2025	1200
21	O059	C091	3/21/2025	45000

Guide Questions

1. Why must dimension tables contain unique keys?

- To ensure each customer is represented only once, enabling accurate filtering and aggregation without duplication.

2. What problems occur if duplicates exist in DimCustomer?

-Double-counting in reports, Incorrect aggregations, Ambiguous relationships, Inconsistent analysis results

Part 3: Define Primary & Foreign Keys

Table	Key	Type
DimCustomer	CustomerID	Primary Key
FactSales	CustomerID	Foreign Key
FactSales	OrderID	Primary Key

Guide Questions

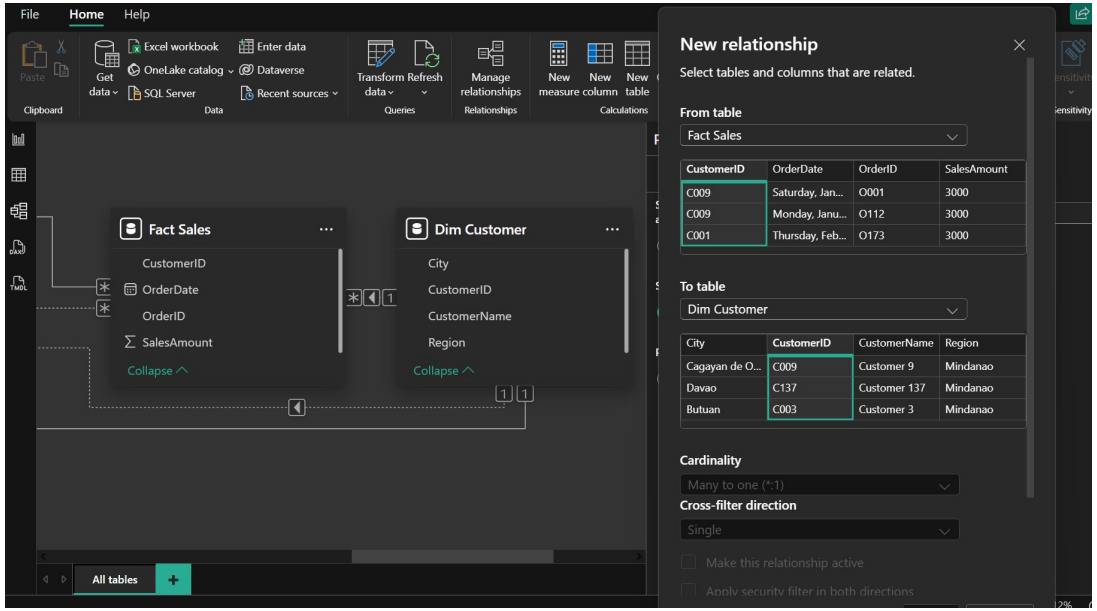
1. What is a primary key?

   - A column (or combination of columns) that uniquely identifies each row in a table. In DimCustomer, CustomerID uniquely identifies each customer.

2. Why is CustomerID a foreign key in FactSales?

   - It references the primary key in DimCustomer, allowing us to connect transaction data to customer attributes (name, city, region).

Part 4: Create Relationships



Guide Questions

1. Why should relationships flow from dimension to fact?
   - Dimensions filter facts (not vice versa) for better performance and intuitive model design. This follows the star schema principle.
2. What happens if the relationship is Many-to-Many?
   - Ambiguous filtering, potential incorrect aggregations, and slower performance. It can cause double-counting issues.

Part 5: Validate the Data Model

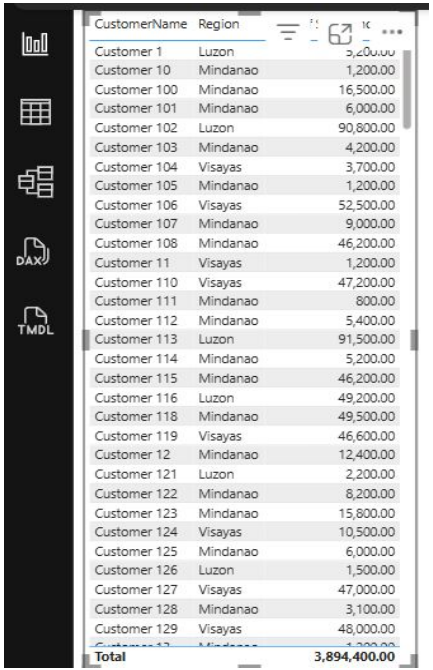
Guide Questions

1. Did sales aggregate correctly per customer?

   - Yes, each customer shows the sum of all their orders.

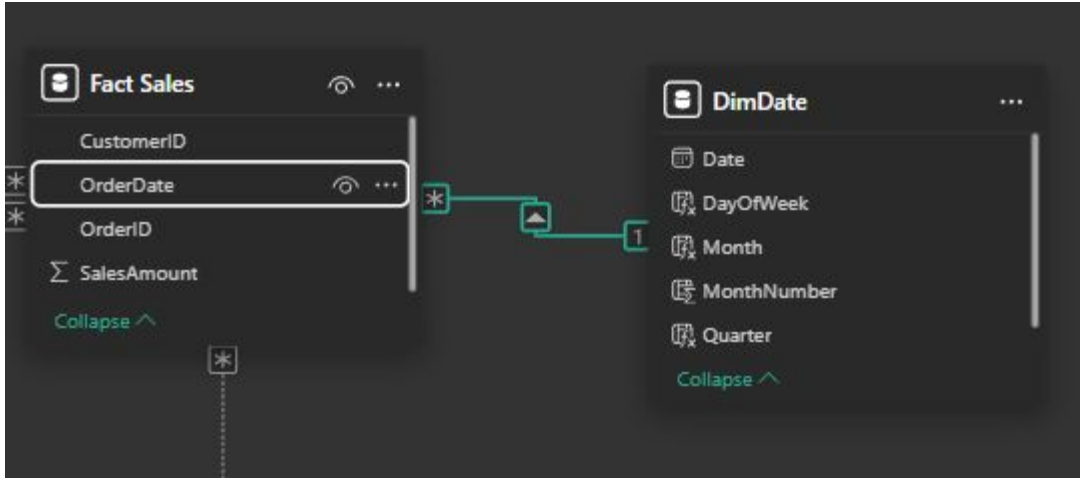
2. What would happen if relationships were missing?

   - Visuals wouldn't show correct data, might show errors, or customers would be disconnected from their sales.



CustomerName	Region	
Customer 1	Luzon	3,200.00
Customer 10	Mindanao	1,200.00
Customer 100	Mindanao	16,500.00
Customer 101	Mindanao	6,000.00
Customer 102	Luzon	90,800.00
Customer 103	Mindanao	4,200.00
Customer 104	Visayas	3,700.00
Customer 105	Mindanao	1,200.00
Customer 106	Visayas	52,500.00
Customer 107	Mindanao	9,000.00
Customer 108	Mindanao	46,200.00
Customer 11	Visayas	1,200.00
Customer 110	Visayas	47,200.00
Customer 111	Mindanao	800.00
Customer 112	Mindanao	5,400.00
Customer 113	Luzon	91,500.00
Customer 114	Mindanao	5,200.00
Customer 115	Mindanao	46,200.00
Customer 116	Luzon	49,200.00
Customer 118	Mindanao	49,500.00
Customer 119	Visayas	46,600.00
Customer 12	Mindanao	12,400.00
Customer 121	Luzon	2,200.00
Customer 122	Mindanao	8,200.00
Customer 123	Mindanao	15,800.00
Customer 124	Visayas	10,500.00
Customer 125	Mindanao	6,000.00
Customer 126	Luzon	1,500.00
Customer 127	Visayas	47,000.00
Customer 128	Mindanao	3,100.00
Customer 129	Visayas	48,000.00
Total		3,894,400.00

Enhancement 1: Add DimDate Table



Enhancement 2: Create a DimRegion Table

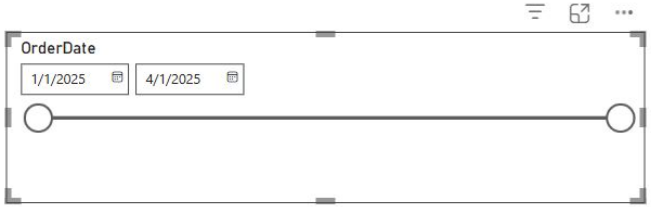
Region	RegionID
Mindanao	REG02
Luzon	REG03
Visayas	REG01

Enhancement 3

Flat Table

City	Luzon	Mindanao	Visayas	Total
Baguio	317,400.00			**317,400.00**
Butuan		468,200.00		**468,200.00**
Cagayan de Oro		230,900.00		**230,900.00**
Cebu			295,500.00	**295,500.00**
Davao		276,700.00		**276,700.00**
Dumaguete			323,300.00	**323,300.00**
Iloilo			422,900.00	**422,900.00**
Legazpi	212,600.00			**212,600.00**
Manila	420,000.00			**420,000.00**
Quezon City	406,900.00			**406,900.00**
Tacloban			269,700.00	**269,700.00**
Total	**1,356,900.00**	**1,226,100.00**	**1,311,400.00**	**3,894,400.00**

Sum of SalesAmount by Region

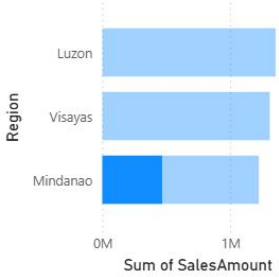


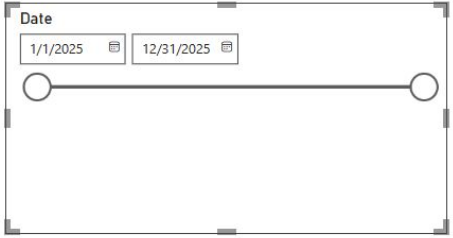
Enhancement 3

Star Schema

Region	Baguio	Butuan	Cagayan de Oro	Cebu	Davao	Dumaguete	Iloilo	Legazpi	Manila	Quezon City	Tacloban	Zamboanga	Total
Luzon	317,400.00							212,600.00	420,000.00	406,900.00			1,356,900.00
Mindanao		468,200.00	230,900.00		276,700.00							250,300.00	1,226,100.00
Visayas				295,500.00		323,300.00	422,900.00				269,700.00		1,311,400.00
Total	**317,400.00**	**468,200.00**	**230,900.00**	**295,500.00**	**276,700.00**	**323,300.00**	**422,900.00**	**212,600.00**	**420,000.00**	**406,900.00**	**269,700.00**	**250,300.00**	**3,894,400.00**

Sum of SalesAmount by Region



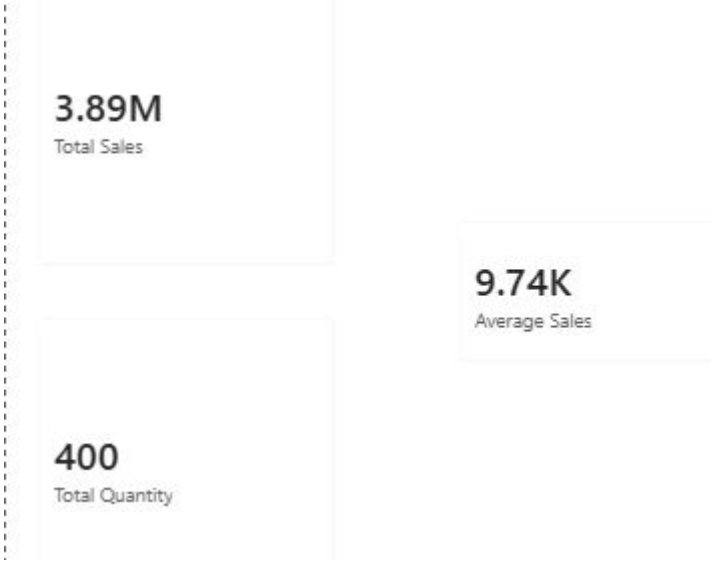


Enhancement 3

Performance Comparison Guide:

Aspect	Flat Table	Star Schema
Data Size	Larger (duplicate customer data)	Smaller (normalized)
Query Speed	Slower	Faster
DAX Complexity	More complex	Simpler
Maintenance	Harder	Easier
Scalability	Poor	Excellent

Introduction to DAX



Guide Questions

Guide Questions

1. What is a DAX measure?

   - A DAX measure is a formula that performs calculations on your data dynamically based on the current filter context

2. Difference between SUM and AVERAGE?

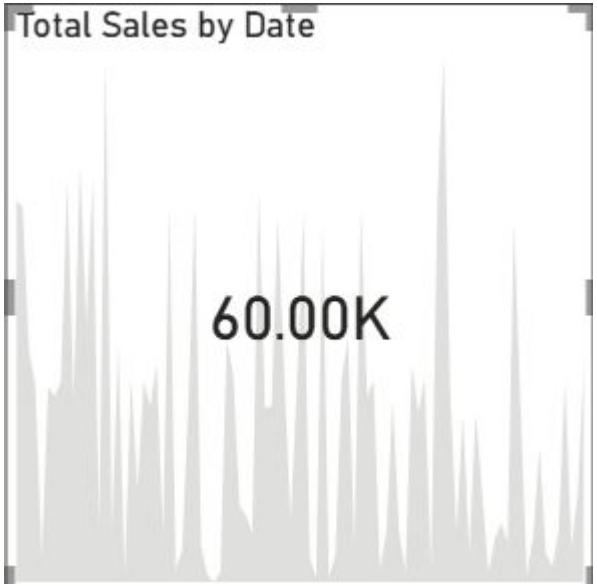
   - Sum adds all values together, Average Calculates arithmetic mean.

3. Why use measures instead of calculated columns?

   - Because measures do calculate results when needed in reports.

Enhancements DAX 1

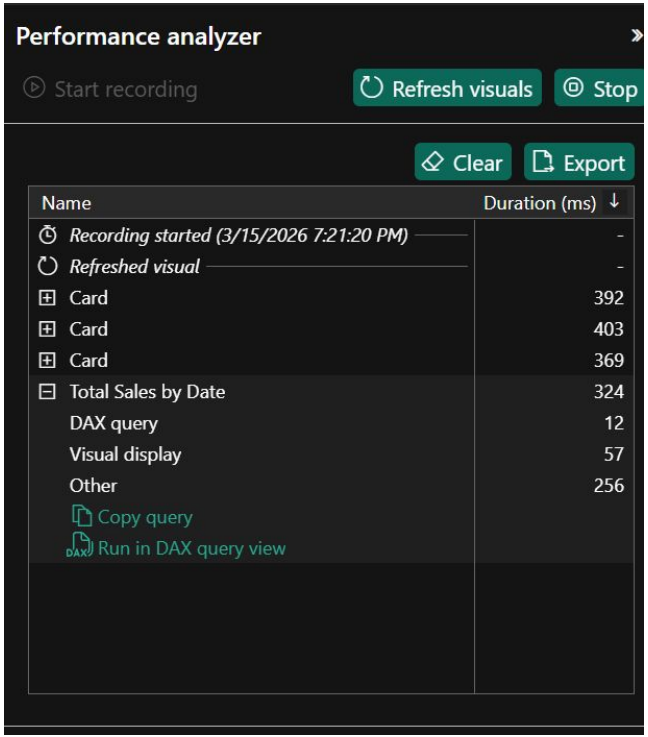
Create a KPI visual using Total Sales.



Enhancement DAX 2

Compare measure vs calculated column

performance.



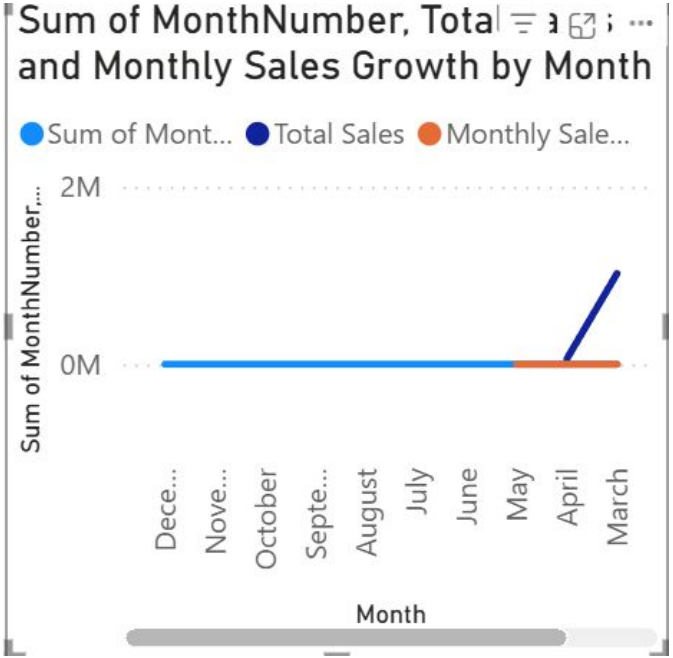
Enhancement DAX 3

Explore COUNT vs COUNTROWS.

DAX Functions

Year to Date

Monthly Sales Growth



Guide Questions

1. What is time intelligence in DAX?
   - Time intelligence in DAX refers to a set of functions that perform calculations across time periods.
2. Why is a date table required?
   - Time intelligence functions need a complete calendar to work correctly. Without it, they get confused by missing dates.
3. How does PREVIOUSMONTH help analyze trends?
   - PREVIOUSMONTH automatically compares this month's sales to last month's sales.
4. What insights can YTD Sales provide?
   - YTD Sales shows your cumulative total from January 1 up to whatever date you're looking at.